MATH 345 Skeleton Notes
Unit 3: Beyond linear maps: local linearization
Section 3.1: General functions and domains

Function types and component functions

Vectors, matrices, and linear maps and Anatomy of a linear map studied maps of the form $\mathbf{x} \mapsto A\mathbf{x}$. We now allow more general rules. The output may be a scalar, a vector, or several scalar component functions collected together.

Definition A $\boxed{\text{vector-valued map}}$ is usually written $F : \mathbb{R}^n \to \mathbb{R}^m$. A $\boxed{\text{scalar-valued function}}$ is usually written $f : \mathbb{R}^n \to \mathbb{R}$. Any vector-valued map $F : \mathbb{R}^n \to \mathbb{R}^m$ has m $\boxed{\text{component functions}}$, or $\boxed{\text{coordinate functions}}$, the scalar-valued functions $F_1, \dots, F_m : \mathbb{R}^n \to \mathbb{R}$ such that

$$F(\mathbf{x}) = \begin{bmatrix} F_1(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{bmatrix}.$$

More generally, vector-valued and scalar-valued functions may only be defined on a subset of vectors in $\mathbb{R}^n$, rather than defined everywhere, so their $\boxed{\text{domain}}$ may be a proper subset of $\mathbb{R}^n$.

In other words, a scalar-valued function takes in an input and outputs *scalars*, and a vector-valued function takes in an input and outputs *vectors* (recall Vector).

Activity

If $\mathbf{x}$ is a point in $\mathbb{R}^n$, recall the length of $\mathbf{x}$, denoted $\|\mathbf{x}\|$, from Length of a vector. The length of a vector defines a scalar-valued function from $\mathbb{R}^n$ to $\mathbb{R}$. Find two different vectors with the same length. What does this say about whether the output determines the input?

Activity

Let A be an $m \times n$ matrix. The linear map (recall Linear map) $T : \mathbb{R}^n \to \mathbb{R}^m$ defined by $T(\mathbf{x}) = T_A(\mathbf{x}) = A\mathbf{x}$ is a vector-valued map. Each component function is a row dot product.
If $\mathbf{a}_1, \dots, \mathbf{a}_m$ are the rows of A, and $T_1, \dots, T_m$ are the component functions of T, then for each j, express the function T_j in terms of the row $\mathbf{a}_j$.

Activity

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Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{bmatrix},$$

which defines a linear map $T_A : \mathbb{R}^3 \to \mathbb{R}^2$. Compute its component functions for a general input $\mathbf{x}$, and its output for the specific input $\mathbf{x} = (3, -1, 2)$.

Activity: Corners are not enough

Vectors, matrices, and linear maps used the unit-square visualization for 2×2 matrices. For nonlinear maps, the corners of a square do not tell the whole story.
Let $T, F : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y \\ y \end{bmatrix}, \qquad F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}.$$

1. Compute the images of the four corners of the unit square under T.
2. Compute the images of the four corners of the unit square under F.
3. Compute the images of $\begin{bmatrix} 0 \\ 1/2 \end{bmatrix}$ under both maps.
4. What does F do to a point $\begin{bmatrix} 0 \\ y \end{bmatrix}$ on the left edge of the unit square?

Note A matrix sends grid lines to grid lines. A nonlinear map can bend grid lines. The square-grid visualization samples points and grid lines, not only the corners.

Activity: Reading a square-grid computation

The following code applies the nonlinear map

$$F\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y^2 \\ y \end{bmatrix}$$

to several input points stored as columns.

```
import numpy as np

def F(P):
    x = P[0, :]
    y = P[1, :]
    return np.vstack([x + y**2, y])

P = np.array([
    [0.0, 1.0, 1.0, 0.0, 0.0],
    [0.0, 0.0, 1.0, 1.0, 0.5],
])

F(P)
```

Output:

```
array([[0.  , 1.  , 2.  , 1.  , 0.25],
       [0.  , 0.  , 1.  , 1.  , 0.5 ]])
```

1. What points are stored in the first four columns of P?
2. What point is stored in the last column of P?
3. What output corresponds to the last column?
4. Why do the first four columns not detect the difference between F and the shear $T(x, y) = (x + y, y)$?
5. Why is a grid more informative than the four corners?

Note: Reading square-grid diagrams A square-grid diagram keeps track of input points and output points at the same time. The color identifies the original input point. The grid lines show how nearby input points move.

For a matrix map, grid lines stay straight and the whole grid is controlled by one matrix. For a smooth nonlinear map, grid lines may bend.

The purpose of the diagram is not to replace computation. It shows what the computation is doing.

The notation $F : \mathbb{R}^n \to \mathbb{R}^m$ includes several cases. If $m = 1$, the output is scalar-valued. If $n = 1$ and $m > 1$, the map traces a curve. If $F(\mathbf{x}) = A\mathbf{x}$, the map is linear. If $F(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, the map is affine.

Domains, ranges, graphs, and level curves

Domain and range

Definition: Domain and range The $\boxed{\text{domain}}$ of a function is the set of all inputs for that function, and the $\boxed{\text{range}}$ is the set of all possible outputs. More formally, if a function f takes inputs from a set X and outputs elements in a set Y, then the domain of f is the set X, and the range is the set of all $y \in Y$ for which $y = f(x)$ for some $x \in X$, which we might write in set notation as $\{y \in Y : y = f(x) \text{ for some } x \in X\}$.

In calculus, we often define the domain of a function f *implicitly*, by defining f in terms of an expression, and then letting the domain of f be the set of all values which, when substituted into that expression, lead to a well-defined quantity. For instance, we might discuss 'the function' $f(x, y) = x/y$, by which we mean the function with domain $\{(x, y) \in \mathbb{R}^2 : y \neq 0\}$ and range $\mathbb{R}$.

For scalar-valued functions of two variables, domains and level curves are subsets of the plane. These pictures help us see where a function is defined and how its outputs change.

Activity

Sketch the domains of the following functions:
$f(x, y) = \ln(y - x^2) + \sqrt{x}$.

Activity

Sketch the domains of the following functions:
$f(x, y) = e^x \sin y$

Activity

Sketch the domains of the following functions:
$f(x, y) = \frac{\sqrt{x}}{\sqrt{y}}$

Graphs and level curves

Definition: The Graph of a Function The $\boxed{\text{graph}}$ of a function f is the set of all pairs $(x, f(x))$, with x in the domain of f.

When f is a scalar-function of two variables, the graph of f is a set of triples $(x, y, f(x, y))$, and thus a subset of $\mathbb{R}^3$, and can therefore be visualized as a surface. See textbook figure fig-8-3-figure-example for a picture of the graph of the function $f(x, y) = xe^{-x^2-y^2}$.

Definition: Level Curves and Contour Plots A $\boxed{\text{level curve}}$ of a scalar function f of two variables is the curve consisting of pairs (x, y) satisfying the equation $f(x, y) = c$, for some $c \in \mathbb{R}$. A $\boxed{\text{contour plot}}$ is a sketch in $\mathbb{R}^2$ depicting level curves of a function f for many different values of c.

The different values of c which are chosen to be depicted in a contour plot are often (but not always) obtained by varying c by a common difference, obtaining 'consecutive' level sets.

Activity

Sketch a contour plot for the function

$$f(x, y) = e^{x^2+y^2}$$

by sketching the level curves $f(x, y) = c$ with $c = 1$, $c = e$, $c = e^4$, and $c = e^9$.